

5(a)(i)	$T = 2.0 \times 10^{-5} \times 6.0 \text{ (= } 1.2 \times 10^{-4} \text{ s)}$	C1
	$f = 1 / (2.0 \times 10^{-5} \times 6.0)$ $= 8300 \text{ Hz}$	A1
5(a)(ii)	new trace shows the same period	B1
	new trace shows amplitude of 10 small squares	B1
5(a)(iii)	(trace is a) vertical line	B1
5(b)(i)	$n\lambda = d \sin \theta$	C1
	$\lambda = (3.4 \times 10^{-6} \times \sin 16^\circ) / 2$ $= 4.7 \times 10^{-7} \text{ m}$	A1
5(b)(ii)	$n = 3.4 \times 10^{-6} (\times \sin 90^\circ) / 4.7 \times 10^{-7} \text{ or } 2 (\times \sin 90^\circ) / \sin 16^\circ$ $(= 7.2 \text{ or } 7.3)$	C1
	highest order = 7	A1